

Beyond Fluency: Diagnosing and Verifying Causal Admissibility in Large Language Models

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ABSTRACT

Evaluation frameworks for LLMs benchmark accuracy and fluency while remaining silent on whether generated explanations satisfy the formal requirements of causal validity, a critical gap in safety-critical deployment. We introduce **Causal-Guard**, a post-hoc neuro-symbolic framework operationalising *causal admissibility* through five Pearl-grounded constraints: temporal precedence (C^1), spatial plausibility (C^2), mechanistic plausibility (C^3), non-spuriousness (C^4), and completeness (C^5). Evaluated across 198 diagnostic scenarios spanning six domains: road maintenance, public events, weather disturbance, traffic accidents, healthcare, and finance, and six LLM families under zero-shot prompting, two systematic failure classes emerge invisible to accuracy-based metrics. First, C^3 failures are near-universal (0.0–2.6% pass rate), establishing that LLMs routinely assert mechanisms contradicting domain physical constraints regardless of scale or architecture. Second, C^2 constitutes a previously undercharacterised failure dimension (47.3–82.8% pass rate), particularly severe in healthcare and finance where spatial plausibility requires non-geographic reasoning. C^1 and C^4 remain robust at $\sim 100\%$, confirming selective rather than global causal deficits. An LLM-as-judge baseline confirms that self-verification systematically misses violations that Causal-Guard’s rule layer catches with 100% recall at sub-100ms latency. Data, code, and knowledge bases are released at (anonymous repo).

Keywords:

causal admissibility, large language models, causal verification, neuro-symbolic AI, LLM evaluation, Pearl’s causal hierarchy

Categories:

Email:

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1 INTRODUCTION

As LLMs are deployed for clinical explanation, infrastructure attribution, and financial risk characterisation, the question of what constitutes a *valid explanation* acquires operational urgency. Current practice assesses correctness and fluency but is structurally silent on a distinct property: whether an explanation satisfies formal causal validity. A fluent explanation asserting a physically impossible mechanism, misattributing causation across spatially disconnected components, or omitting necessary causal factors is causally inadmissible and its deployment in safety-critical settings carries direct risk of consequential misallocation.

Existing benchmarks do not address this gap. Capability assessments such as CLadder [16] and e-CARE [2] ask: can the model reason causally? Faithfulness evaluations [13][4] ask: does the explanation reflect the model’s computation? Neither asks: is the explanation causally valid under domain facts? These are orthogonal questions. Formally, $P(\hat{y}=y|x,M) \perp \alpha(E,S)$ in general, where $\alpha(E,S) = \models_i[E \models C_i|S]$ denotes causal admissibility. Existing evaluation leaves $\alpha(E,S)$ entirely unmeasured. Evaluation frameworks that conflate these questions leave a systematic, consequential failure class undetected.

We introduce **Causal-Guard**, a post-hoc neuro-symbolic framework formalising causal admissibility as a verifiable property. Causal-Guard operationalises admissibility through five constraints grounded in Pearl’s SCM framework [11], spanning all three levels of Pearl’s causal hierarchy. The framework requires no model internals, fine-tuning, or generation-time intervention, making it compatible with any API-accessible or locally deployed LLM. Evaluated across 198 scenarios and six domains, the results establish causal admissibility failure as a systematic cross-domain property of current LLM-generated explanations. This paper makes four contributions: (1) five

Pearl-SCM-grounded constraints with formal definitions and auditable checker implementations; (2) a reproducible multi-domain evaluation protocol ($\kappa \geq 0.81$); (3) empirical characterisation across six LLM families establishing near-universal C^3 failure and substantial C^2 failure; and (4) an automated verification system with 100% recall on physical-impossibility violations at 73ms mean latency.

2 RELATED WORK

2.1 LLM Evaluation and Faithfulness

MMLU [3], GSM8K [1], and BIG-Bench [14] measure answer correctness $P(\hat{y}=y|x,M)$. Mondorf and Plank [8] document systematic dissociation between accuracy and reasoning quality. Turpin et al. [13] show CoT explanations frequently diverge from actual model computation; Jacovi and Goldberg [4] formalise faithfulness as $\rho(\varphi(E), h(M,x)) > \tau$. Neither framework addresses the structural question of causal validity: a model can achieve high $P(\hat{y}=y)$ while generating causally inadmissible explanations, and can produce faithful accounts of its own (flawed) causal reasoning.

2.2 Causal Reasoning in LLMs

Kiciman et al. [7] show LLMs conflate $P(Y|X)$ with $P(Y|\text{do}(X))$ without explicit structure. Jin et al. [6] report near-chance performance on interventional queries across multiple model families. Wang et al. [15] show self-consistency improves reliability but does not address structural verification against domain facts. These studies are capability diagnostics: they measure whether a model can reason causally on synthetic controlled inputs. They provide no taxonomy of deployment violation types, no formal characterisation of how violations occur, and no automated verification mechanism. The capability question and the admissibility question are orthogonal; answering the former does not answer the latter.

2.3 Neuro-Symbolic Verification

Logic-LM [10] augments generation with symbolic solvers, requiring model access or generation-time intervention, which precludes black-box API

compatibility. Saparov and He [12] propose constraint-based evaluation of CoT reasoning against propositional logical principles, the closest structural precursor to Causal-Guard but limited to propositional logic, do not address causal structure, and provide no deployable verification system. Causal-Guard differs in three respects: post-hoc operation without model access, grounding in Pearl’s SCM rather than propositional logic, and a deployable pipeline with measured 73ms mean latency evaluated across six domains.

2.4 Causal Explanation Benchmarks

CLadder [16] evaluates causal reasoning over symbolic controlled inputs; e-CARE [2] benchmarks causal explanation with human reference labels. Both target capability under laboratory conditions. Neither addresses admissibility as a formal automatable property nor provides a verification system. The LLM-as-judge paradigm [17] fails for causal verification because the same defective world model that generates a causally inadmissible explanation accepts it under evaluation, confirmed empirically in Section 5.4.

2.5 Positioning Summary

Table 1 Positioning of Causal-Guard relative to prior work.

Work	Pearl	Post-hoc	Multi-dom.	Auto-verify	Viol.tax.
MMLU/GSM8K	X	✓	✓	✓	X
Mondorf & Plank [8]	X	✓	✓	X	X
Turpin et al. [13]	X	✓	Part.	X	X
Kiciman et al. [7]	Part.	✓	X	X	X
Jin et al. [6]	✓	✓	X	X	X
CLadder [16]	✓	Part.	X	X	X
e-CARE [2]	X	Part.	X	X	X
Saparov & He [12]	X	Part.	X	X	Part.
Logic-LM [10]	X	X	X	✓	X
LLM-as-judge [17]	X	✓	✓	✓	X
Causal-Guard	✓	✓	✓	✓	✓

3 THE CAUSAL-GUARD FRAMEWORK

3.1 Formal Definitions

Let S be a scenario with domain facts $\mathcal{FS}$ and ground-truth minimal factor set $F_{\min} \subseteq \mathcal{FS}$. Let E be an LLM explanation parsed into causal triples $T(E) = \{(ck, rk, ek)\}$ where ck is the asserted cause, $rk \in \{\text{caused, led to, resulted in, due}$

to, triggered} is the causal connective, and e_k is the asserted effect. E is *causally admissible* w.r.t. S if:

$$Admissible(E, S) = 1 \Leftrightarrow \forall C_i \in C: T(E) \models C_i | S$$

This is a strict conjunction: a single violation renders E inadmissible.

Grounded in Pearl’s SCM [11], an admissible explanation is a valid subgraph of the true causal DAG G^* over domain variables:

$$Admissible(E, S) = 1 \Rightarrow G(E) \subseteq G^* \wedge F_p^{on} \subseteq V(G(E))$$

Constraints C^1 – C^4 enforce the subgraph containment condition (no false edges); C^5 enforces the completeness condition (no missing necessary factors).

3.2 Constraint Hierarchy

C^1 and C^2 operate at Pearl’s associational level and are verifiable from $\mathcal{FS}$ alone. C^3 and C^4 operate at the interventional level, requiring knowledge of domain mechanisms beyond what is observable in $\mathcal{FS}$. C^5 operates at the counterfactual level, requiring reasoning about factors necessary for the outcome under perturbation, the most demanding epistemological requirement and the one most sensitive to knowledge base completeness. This hierarchy directly determines checker design: associational constraints use rule-based or network lookup approaches; interventional constraints require embedding similarity against an external KB plus hard-rule overrides; the counterfactual constraint requires synonym-expanded coverage of domain-specific minimal-factor templates.

3.3 Constraint Formalisations

Table 2 Constraint specifications with domain-conditioned interpretations and checker performance (n=41).

Constraint	Formal condition	Pearl level	Checker	Prec./Rec.	ms
C_1 Temporal	$t^c < t^e$	Assoc.	Rule	100/100	12±3
C_2 Spatial	$\text{dist} \leq \theta_{sp}$	Assoc.	Network	100/100	15±4
C_3 Mechanism	$\text{Sim} \geq \theta_m \wedge \varphi = 1$	Interv.	Emb.+Rule	94/100†	28±6
C_4 Non-spur.	$\text{justif} > \alpha \cdot \text{corr}$	Interv.	Regex	100/100	8±2
C_5 Complete	$ F_p \cap F_i / F_p \geq \beta$	Countf.	Sem.+syn.	66/66	10±3
Pipeline total					73±18

† C^3 precision/recall: physical-impossibility subclass only. Overall C^3 precision at $\theta_m=0.4$: 21.1%. See §6.2.

C₁ Temporal precedence.

$C^1(c,e,S) = 1[tc < te]$. Rule-based timestamp extraction covering absolute dates, relative temporal markers, and ordinal sequencing. Precision and recall: 100%.

C₂ Spatial plausibility.

$C^2(c,e,S,D) = 1[\text{dist}_{N^D}(L^c, L^e) \leq \theta \text{sp}^D]$. Network N^D is domain-specific: geographic road graph for transportation ($\theta=2\text{km}$); anatomical adjacency graph encoding organ-system connectivity and pharmacological pathways for healthcare; market linkage graph encoding sector co-movement relationships for finance. Non-geographic spatial semantics explain substantially lower C^2 pass rates in the expanded evaluation.

C₃ Mechanistic accuracy.

$C^3(c,e,S,D) = 1[\max_{p \in P} \text{Sim}(M,p) \geq \theta_m] \wedge \varphi(M,S)$. Stage 1: cosine similarity against 42-entry Mechanism KB (all-MiniLM-L6-v2, $\theta_m=0.4$). Stage 2: hard-rule overrides for physical impossibility: ice formation at $T > 0^\circ\text{C}$; drug affecting tissue without delivery pathway; price impact without execution channel. Rule layer: 100% recall on physical-impossibility violations, 94.3% precision. Overall precision at $\theta_m=0.4$: 21.1%; deliberate safety-first calibration.

C₄ Non-spuriousness.

$C^4(c,e,S) = 1[\text{justif}(c,e) > \alpha \cdot \text{corr}(c,e)]$, $\alpha=1.0$. Regex matching against 12-entry spurious-correlation library extended with domain-specific patterns for healthcare and finance.

C₅ Completeness.

$C^5(E,S) = 1[|F_{\min} \cap FE| / |F_{\min}| \geq \beta]$, $\beta=0.6$. Synonym-expanded matching against 16 incident-type templates extended with domain-specific families for healthcare (symptom-pathway-outcome triples) and finance (trigger-transmission-impact triples). F1: 66.1%; false negatives arise from synonym coverage gaps in novel domain terminology.

3.4 Pipeline Architecture

Stage 1 (claim extraction): dependency parse + connective matching → triples. Connective vocabulary is domain-extended with clinical and financial terminology. **Stage 2** (constraint checking): five checkers execute in parallel, each accessing only its designated KB, fully modular and independently auditable. **Stage 3** (admissibility judgment): conjunction over all checkers; first failing triple per constraint logged with ID, triple, and violation type. **Stage 4** (audit logging): all inputs, checker outputs, and verdicts written to PostgreSQL with versioned KB snapshots. Post-hoc compatible with any API-accessible or locally deployed LLM.

4 EVALUATION METHODOLOGY

4.1 Scenario Construction

Each scenario satisfies four properties: (1) known ground truth: F_{min} specified by domain annotation; (2) controlled variation across causal complexity, distractor presence, and ambiguity; (3) naturalism: reflecting realistic deployment conditions; and (4) targeted testing: at least one constraint tested and at least one physically impossible C^3 variant included. The benchmark comprises 198 diagnostic scenarios across six safety-critical domains: road maintenance, public events, weather disturbance, traffic accidents, healthcare, and finance.

4.2 Four-Stage Violation Detection

Stage 1: Claim extraction. LLM explanations are parsed into (cause, relation, effect) triples via dependency parsing and connective matching. **Stage 2: Constraint checking.** Each triple is evaluated against all five constraints in parallel using ground-truth facts and domain KBs. **Stage 3: Admissibility judgment.** One violation → inadmissible; constraint ID, offending triple, and violation type are logged. **Stage 4: Annotation.** ≥ 2 independent annotators; $\geq 20\%$ double-annotation; $\kappa \geq 0.81$ across all domains (achieved $\kappa = 0.84$ transportation, $\kappa = 0.81$ deforestation).

5 EMPIRICAL VALIDATION

5.1 Pilot Evaluation: Urban Transportation

We first evaluated Causal-Guard on 41 structured scenarios in the urban transportation domain (Llama 3.3 70B, zero-shot). 52.5% of generated explanations violate at least one constraint despite uniformly high fluency, a violation rate invisible to standard metrics. C^5 is the dominant failure at 33%, reflecting systematic omission of causally necessary factors from F_{min} . C^3 follows at 20%, concentrated in weather-disruption scenarios requiring thermodynamic constraint satisfaction. C^1 , C^2 , and C^4 show zero violations, confirming robust pre-training acquisition of temporal, geographic-spatial, and correlational structure. The zero C^2 rate reflects geographic homogeneity of pilot scenarios, all locations from a single urban road network a confound resolved by the expanded multi-domain evaluation.

Table 3 Pilot study violation rates (Llama 3.3 70B, $n=41$). C_1 , C_2 , C_4 = 0% throughout.

Category	n	C_3 viol.%	C_5 viol.%	Any%
Weather disruptions	10	50%	50%	80%
Traffic accidents	10	30%	30%	50%
Road maintenance	10	0%	20%	20%
Public events	10	0%	30%	30%
Overall	41	20%	33%	52.5%

5.2 Mechanistic Failure: Controlled Temperature Experiment

Scenario WD-05 presented black ice formation at three temperatures: -2°C (physically valid), 2°C and 5°C (both impossible). Llama 3.3 70B, Gemini 2.0 Flash, and Gemini 3 Pro all correctly identified black ice at -2°C . All three accepted the physically impossible claim at 2°C and 5°C without qualification. Gemini 3 Pro exhibited a 45.9-second reasoning delay before the 5°C response; sufficient time to apply the freezing-point constraint yet produced an explanation functionally identical to faster models in the same condition. We term this pattern **causal sycophancy** within the C^3 failure class: propagation of a physically impossible mechanism from the prompt despite demonstrable possession of the relevant domain knowledge. This is a verification failure, not a knowledge deficit. Next-token training provides no gradient signal for detecting constraint violations between prompt-embedded claims and world knowledge;

the model is optimised to continue a coherent narrative, not audit it against physical constraints.

5.3 Primary Multi-Domain Evaluation

We expanded to 198 diagnostic scenarios across all six safety-critical domains. Six LLM families were evaluated under zero-shot prompting: Llama 3.3 70B, GPT-OSS 120B, Qwen 32B, Mistral Small 4 119B, Llama 3.1 405B-Instruct, and DeepSeek-R1-Distill-Qwen-7B. Scenario counts vary ($n=184\text{--}195$) due to generation failures and API timeouts excluded per model and documented in the released annotation data.

Table 4 Per-constraint pass rates across six LLM families. Higher is better.

Model	C ₁	C ₂	C ₃	C ₄	C ₅	n
Llama 3.3 70B	100	78.4	2.6	100	100	194
GPT-OSS 120B	100	48.7	1.6	100	100	191
Qwen 32B	100	51.1	1.6	99.5	100	184
Mistral Small 4 119B	100	47.3	0.5	100	100	184
Llama 3.1 405B	100	82.8	0.0	100	100	186
DeepSeek-R1-D-7B	100	80.5	0.0	100	100	195

C¹ and C⁴ robustness. Near-perfect pass rates confirm that temporal precedence and non-spuriousness are structural properties reliably acquired from pre-training. The single exception: Qwen 32B C⁴ at 99.5% corresponds to an ambiguous financial scenario where the spurious-correlation boundary is genuinely contested in the domain literature.

C⁵ resolution. The dominant pilot failure (33%) reverses to $\sim 100\%$ pass in the expanded evaluation. This reflects two implementation changes: (i) the synonym lexicon was extended with clinical, financial, and event-domain terminology absent from the transportation-only original; (ii) Fmin annotation was performed with more precisely specified factor sets. These changes address the checker’s synonym coverage gap rather than reflecting genuine model improvement.

C³ near-universal failure. Mechanistic accuracy collapses to 0.0–2.6% pass rate across all models. This is scale-invariant (Llama 3.1 405B: 0.0%; 7B distill: 0.0%), architecture-invariant (dense, MoE, distilled reasoning), and domain-invariant across all six domains. The universality establishes C³ inadmissibility as a systematic property of current LLMs under zero-shot prompting.

C² variance. The 35.5 percentage-point range (47.3–82.8%) is partially explained by scale: the two largest models achieve the highest C² pass rates, and domain distribution: C² failures concentrate in healthcare and finance, where spatial plausibility is defined over anatomical and market linkage graphs, domains underrepresented in pre-training corpora relative to geographic co-occurrence data.

5.4 LLM-as-Judge Baseline

We applied two LLM-as-judge configurations to the 41 transportation scenarios: Llama 3.3 70B self-judge and GPT-4o independent judge. Both were provided $\mathcal{FS}$ and $\mathcal{E}$ under zero-shot evaluation prompts instructing them to identify causal constraint violations.

Table 5 LLM-as-judge vs. Causal-Guard (n=41).

Method	C ₃ rec.	C ₅ F ₁	Notes
Causal-Guard	100%	66.1%	Neuro-symbolic; rule + embedding
Judge: Llama 3.3 70B	20%	31.2%	Self-judge; misses 80% of C ₃ violations
Judge: GPT-4o	40%	44.8%	Independent judge; misses 60% of C ₃ violations

Both judges accept physically impossible claims that appear confidently in evaluation text, the same next-token objective that produces inadmissible claims drives the judge toward narrative acceptance rather than constraint verification. The Causal-Guard rule layer achieves 100% C³ recall through explicit constraint composition against the KB, independent of claim surface confidence. Symbolic verification is structurally necessary for this failure class, not substitutable by a more capable or independent LLM judge.

5.5 Study 2: Deforestation Attribution

To assess generalisation beyond transportation, we evaluated Causal-Guard on 60 deforestation attribution scenarios (Gemma 3 4B) under four constraint enforcement levels S0–S3. S0 produced 40 violations; S3 reduced this to 1 case, a 97.5% reduction. Dominant S0 failures temporal lag neglect (C¹), trigger-as-cause misattribution (C³), geometric red-herring bias (C²), map directly onto the unified constraint taxonomy, confirming generalisation without modification. Inter-rater $\kappa=0.81$. The near-complete elimination under S3 establishes that causal admissibility failures are addressable, not irreducible.

5.6 Cross-Study Synthesis

Table 6 Cross-study summary.

Dimension	Pilot: transport	Primary: multi-domain	Study 2: deforestation
Domains	Urban transport	6 domains	Environmental science
Models	Llama 3.3 70B	6 frontier families	Gemma 3 (4B)
Scenarios	41	~198	60
Primary viol.	C_3, C_5	C_3 (universal), C_2	C_1, C_3, C_2
Inter-rater κ	0.84	0.81–0.84	0.81
Key finding	C_3/C_5 prevalent	C_3 scale/arch invariant	97.5% reduction under S3

The three studies are analytically complementary. The pilot establishes baseline violation prevalence and calibrates the detection pipeline in a controlled single-domain setting. The primary multi-domain evaluation establishes that C^3 inadmissibility is scale-invariant, architecture-invariant, and domain-invariant, a finding underivable from the pilot alone. The deforestation study establishes that violations are addressable through constraint enforcement, closing the diagnostic loop. Together the studies characterise causal admissibility failure as (i) prevalent under zero-shot prompting, (ii) systematic and constraint-specific rather than random, and (iii) substantially reducible through explicit constraint enforcement.

6 DISCUSSION

6.1 Why C_3 Fails Universally

The next-token training objective optimises $P(x_t|x<t)$, with gradient $\nabla\theta l = -E[\nabla\theta \log P(x_t|x<t)]$ orthogonal to physical constraint satisfaction. The model receives no penalisation signal for restating a physically impossible prompt claim. The WD-05 experiment is precise: correct knowledge at -2°C but failure to compose it with a prompt-supplied 5°C demonstrates that the deficit is compositional, not knowledge-based. The C^3 rule layer performs exactly this composition. The universality across scale, architecture, and domain: including distilled reasoning models (DeepSeek-R1) explicitly trained for multi-step reasoning confirms that the failure is structural to the training objective, not remediable through scale or architectural modification alone.

6.2 Why C_2 Shows Model-Level Variance

Unlike C^3 , C^2 shows a 35.5pp range. The two largest models achieve the highest pass rates, consistent with scale providing marginal benefit for associational-level spatial reasoning. C^2 failures concentrate in healthcare and finance, where spatial plausibility is defined over anatomical and market linkage graphs, domains underrepresented in pre-training corpora relative to geographic co-occurrence data. Even the best-performing model reaches only 82.8%, indicating that non-geographic spatial plausibility constitutes a genuine capability gap rather than purely a verification failure.

6.3 C_3 Precision and Deployment Guidance

At $\theta_m=0.4$, overall C^3 precision is 21.1% (~ 4 false alarms per true violation). We recommend a two-tier protocol: (1) **Hard violations** (rule layer: physical impossibility) trigger immediate rejection or mandatory review; 100% recall, 94.3% precision. (2) **Embedding flags** ($\theta=0.4$), route to human triage; raising θ to 0.5–0.6 yields 31.6–42.9% precision at 75–92% recall, appropriate for lower-stakes or higher-throughput settings. Domain calibration of θ from a gold-standard set is required before production deployment.

6.4 Limitations

Knowledge base completeness. $\sim 15\%$ of novel-domain C^3 violations would be missed due to KB gaps; C^5 synonym coverage is a further limitation. Community extension via the open release is the primary mitigation. **Annotation scalability.** $\kappa \geq 0.81$ does not scale automatically to new domains; KB development and threshold calibration are required per domain. **Evaluation scope.** All evaluations use zero-shot prompting; few-shot constraint-grounded prompting may partially mitigate C^3 and C^2 failures and is deferred to future work. **English-only.** Multilingual extension via translation preprocessing is a natural next step.

7 BENCHMARK AND SYSTEM RELEASE

Causal-Guard is released at <https://github.com/anonymous/Causal-Guard> (anonymised for review). The release includes: (1) 198 annotated diagnostic scenarios with F_{min} ground truth and decision-tree annotation guidelines for all

five constraints across six domains; (2) JSON knowledge base schemas, Mechanism KB (42 entries), Spurious Patterns Library (12 domain-extended entries), Completeness Templates (16 incident-type families with domain-specific synonym expansions); (3) complete pipeline code, PostgreSQL audit schema, and API integration examples compatible with open-weight and commercial black-box models; (4) raw annotation data, inter-rater reliability materials, and per-model generation exclusion logs.

8 CONCLUSION

We presented **Causal-Guard**, a constraint-based neuro-symbolic framework formalising causal admissibility through five Pearl-grounded constraints with automated verification at sub-100ms latency. Evaluated across 198 scenarios, six domains, and six LLM families, the framework establishes two primary findings: (1) C^3 (mechanistic accuracy) failure is near-universal, scale-invariant, architecture-invariant, and domain-invariant a structural consequence of the next-token training objective; and (2) C^2 (spatial plausibility) constitutes a co-equal and previously undercharacterised failure dimension, severe in non-geographic domains. C^1 and C^4 robustness across all models confirms that causal reasoning deficits are selective, pointing toward targeted mitigation strategies. Trustworthy AI in safety-critical domains requires evaluation frameworks that test whether outputs are causally valid, not merely correct and verification systems that enforce validity before outputs reach decision-makers. Causal-Guard provides both. Future work will develop fine-tuned violation classifiers, extend to multilingual settings, and investigate constraint-grounded few-shot prompting as a generation-time mitigation for C^3 failures.

1 REFERENCES

- [1] Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., & Schulman, J. (2021). Training verifiers to solve math word problems. arXiv preprint arXiv:2110.14168. <https://doi.org/10.48550/arXiv.2110.14168>
- [2] Du, L., Ding, X., Xiong, K., Liu, T., & Qin, B. (2022). e-CARE: A new dataset for exploring explainable causal reasoning. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics

- (Volume 1: Long Papers), pp. 432–446.
<https://doi.org/10.18653/v1/2022.acl-long.33>
- [3] Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., & Steinhardt, J. (2021). Measuring massive multitask language understanding. In Proceedings of the 9th International Conference on Learning Representations (ICLR 2021).
<https://doi.org/10.48550/arXiv.2009.03300>
 - [4] Jacovi, A., & Goldberg, Y. (2020). Towards faithfully interpretable NLP systems: On the concept of explanations. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pp. 4403–4412. <https://doi.org/10.18653/v1/2020.acl-main.386>
 - [5] Garcez, A. S., Lamb, L. C., & Gabbay, D. M. (2022). Neural-symbolic learning and reasoning: The 3rd wave. arXiv preprint arXiv:2012.05876. <https://doi.org/10.48550/arXiv.2012.05876>
 - [6] Jin, Z., Budhiraja, A., & Sachan, M. (2024). Evaluating the interventional reasoning capabilities of large language models. arXiv preprint arXiv:2404.05545. <https://doi.org/10.48550/arXiv.2404.05545>
 - [7] Kiciman, E., Ness, R., Sharma, A., & Tan, C. (2023). Causal reasoning and large language models: Opening a new frontier for causality. arXiv preprint arXiv:2305.00050.
<https://doi.org/10.48550/arXiv.2305.00050>
 - [8] Mondorf, P., & Plank, B. (2024). Beyond accuracy: Evaluating the reasoning behavior of large language models — a survey. arXiv preprint arXiv:2404.01869. <https://doi.org/10.48550/arXiv.2404.01869>
 - [9] Pan, L., Saxon, M., Xu, W., Nathani, D., Wang, X., & Wang, W. Y. (2023). Automatically correcting large language models: Surveying the landscape of diverse self-correction strategies. arXiv preprint arXiv:2308.03188. <https://doi.org/10.48550/arXiv.2308.03188>
 - [10] Pan, L., Albalak, A., Wang, X., & Wang, W. Y. (2023). Logic-LM: Empowering large language models with symbolic solvers for faithful logical reasoning. arXiv preprint arXiv:2307.07696.
<https://doi.org/10.48550/arXiv.2307.07696>
 - [11] Pearl, J. (2009). Causality: Models, Reasoning, and Inference (2nd ed.). Cambridge University Press.
 - [12] Saparov, A., & He, H. (2023). Language models are greedy reasoners: A systematic formal analysis of chain-of-thought. In Proceedings of the 11th International Conference on Learning Representations (ICLR 2023). <https://doi.org/10.48550/arXiv.2210.01240>
 - [13] Turpin, M., Michael, J., Perez, E., & Bowman, S. R. (2024). Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. In Advances in Neural Information Processing Systems 36 (NeurIPS 2023).
<https://doi.org/10.48550/arXiv.2305.04388>
 - [14] Srivastava, A., Rastogi, A., Rao, A., et al. (2022). Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. arXiv preprint arXiv:2206.04615.
<https://doi.org/10.48550/arXiv.2206.04615>
 - [15] Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., & Zhou, D. (2023). Self-consistency improves chain of thought

- reasoning in language models. In Proceedings of the 11th International Conference on Learning Representations (ICLR 2023).
<https://doi.org/10.48550/arXiv.2203.11171>
- [16] Zevcevic, M., Willig, M., Dhimi, D. S., & Kersting, K. (2023). CLadder: Assessing causal reasoning in language models. In Advances in Neural Information Processing Systems 36 (NeurIPS 2023).
<https://doi.org/10.48550/arXiv.2312.04350>
- [17] Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., & Stoica, I. (2023). Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In Advances in Neural Information Processing Systems 36 (NeurIPS 2023).
<https://doi.org/10.48550/arXiv.2306.05685>
- [18] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. In Advances in Neural Information Processing Systems 35 (NeurIPS 2022).
<https://doi.org/10.48550/arXiv.2201.11903>